

STOCK WATCH

Research. Analyse. Forecast.

Windows desktop application | Project summary | September 2026

PURPOSE & PROBLEM ADDRESSED

Stock Watch brings Indian equity research into one local workspace. It helps users review prices, financial performance, technical indicators and model-based projections without switching between separate research tools.

KEY CAPABILITIES

Research dashboard: searchable NSE company directory, NIFTY 50 and SENSEX quotes, market status, company overview, peer comparison and linked news.

Analysis: interactive Plotly price and volume charts, moving averages, RSI, MACD, ATR and Bollinger Bands; financial trends and valuation/health calculations where inputs are available.

Screening & reporting: a separate filterable NSE screener, light/dark themes, Excel exports and a two-page projection PDF.

FORECASTING & TRACKING

Bull and Bear scenarios use observed historical return paths or moving-block bootstrap simulations for longer horizons. Probabilities total 100%, with the higher-probability scenario shown first. Calculated stop-loss levels use volatility and support inputs. Immutable pinned projections are stored locally and tracked against actual prices.

TECHNOLOGY & DATA

Python handles data retrieval, calculations and SQLite storage. HTML, CSS and JavaScript provide the interface; Plotly.js renders charts inside PySide6 QtWebEngine. Sources include Yahoo Finance/yfinance, official NSE/BSE feeds, reported financial-summary data and news RSS. Cached data and verified alternate-listing history improve coverage; corporate-action treatment is disclosed.

DELIVERY & VALIDATION

Delivered as a PyInstaller Windows EXE with its accompanying runtime folder; Python installation is not required. Packaging tests passed for startup, company search, charts, forecasts, navigation, saved-projection persistence, screener queries, themes and PDF/Excel exports. GitHub distribution is split into three checksum-verified files, each below 100 MB.

LIMITATIONS & INTENDED USE

For research and education. Quotes may be delayed and depend on provider availability. Missing inputs remain unavailable. Forecast probabilities are uncalibrated historical/simulated frequencies, not guarantees. Fresh data requires internet access; user preferences, saved projections and tracking records remain on the local PC.